

KEY WINDOWS SERVER ADMINISTRATION TERMS

1. Active Directory (AD)

Definition:

Active Directory is a centralized system used in Windows Server to manage users, computers, printers, and other resources on a network.

It helps administrators organize and secure the entire system efficiently.

Real-Life Example (Padri Vjeko Centre):

Imagine Active Directory as the **main school record office** at Padri Vjeko Centre. It keeps all the information about **students, teachers, classrooms, departments, and school rules** in one place, so the system manager can easily manage and control everything.

2. Domain Controller (DC)

Definition:

A Domain Controller is a server that stores the Active Directory database and handles login requests, authentication, and access control for all users in the network.

Real-Life Example:

Think of a Domain Controller as the **school gate security office**. When a student or teacher arrives (logs in), the security officer checks their **ID card (username and password)** against the school's records (Active Directory) before allowing them inside.\

3. Forest

Definition:

A forest is the **highest level** in the Active Directory structure that contains one or more domains that trust each other and share a common schema.

Real-Life Example:

Imagine the forest as the **entire education system** managed by Padri Vjeko Centre TSS – if the school later opens new branches (like Padri Vjeko Centre Kigali and Padri Vjeko Centre Butare), all these branches could be separate domains within one forest.

4. Tree

Definition:

A tree is a **collection of related domains** that share a common namespace (for example, **nit.padrivjeko.local, fad.padrivjeko.local**).

Real-Life Example:

Each department (NIT, FAD, WOT, BDC) could be a **tree** under the main school domain. They are connected but have their own structure.

5. Organizational Units (OUs)**Definition:**

Used to organize users and computers into manageable groups within a domain.

Example:

Each department (NIT, FAD, WOT, BDC) is an OU, and within each, there could be sub-OUs like **Teachers**, **Students**, and **Lab Computers**.

6. Global Catalog (GC)**Definition:**

A distributed data repository that provides information about every object in the directory.

Example:

It's like having a **summary book** listing all students and teachers from every department, so anyone can find contact information quickly.

7. Trust Relationship**Definition:**

A trust relationship allows users from one domain to access resources in another domain securely.

Real-Life Example:

If the Padri Vjeko Centre TSS (main campus) wants its users to access resources from the Padri Vjeko Centre Kigali branch, a **trust** is created between the two schools.

8. Group Policy (GPO)**Definition:**

Rules applied to users or computers in a domain to control configurations and security settings.

Example:

A rule like "All computers must lock after 10 minutes of inactivity" or "Students cannot install new software" is set through GPO.

9. User Accounts

Definition:

Accounts created for each individual to log in and access network resources.

Example:

Every student and teacher at Padri Vjeko Centre has a **unique login ID** (like `john.nit@padrivjeko.local`).

10. Computer Accounts**Definition:**

Each computer in the network has an account in AD so the system can manage and secure it.

Example:

The **Computer Lab PCs** (e.g., LabPC01, LabPC02) each have their own accounts in AD to allow monitoring and control.

11. Security Groups and Distribution Groups**Definition:**

- **Security Groups:** Used to assign permissions to many users at once.
- **Distribution Groups:** Used for sending emails to many users.

Example:

- A **Security Group** called “NIT_Students” may allow access to NIT lab computers only.
- A **Distribution Group** “AllTeachers@padrivjeko.local” sends an announcement email to all teachers.

12. Schema**Definition:**

The schema defines the **structure and types of data** stored in Active Directory – for example, what information a student or teacher record contains.

Example:

Just like a **student registration form** defines what details are needed (name, reg number, department, year), the schema defines what fields every object in AD must have.